MIL-P-36637B
3 June 1976
SUPERSEDING
MIL-P-36637A
16 September 1971

MILITARY SPECIFICATION

PIPETTING MACHINE, SEMI-AUTOMATIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two sizes of manually operated pipetting machines.

1.2 Classification.

* 1.2.1 Sizes. The pipetting machines shall be the following sizes:

Size 1 - 0 to 10 milliliters (ml.) range (30 ounces).
Size 2 - 0 to 1 ml. range (16 ounces).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Military

MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-L-10547- Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as herein specified. All interior surfaces which come into contact with reagents during use shall be fabricated of glass, plastic or aluminum.

3.2 Style and design. The pipetting machine shall consist of a bottle, cap, vent and measuring and dispensing assembly and shall be in accordance with Figures 1 or 2.

3.2.1 Bottles. Bottles shall be fabricated of glass. Neck shall be threaded for easy removal and closure of the cap. Color of bottle shall be uniform for each contract.

3.2.2 Cap. Cap shall be fabricated of plastic and shall be threaded to mate with the neck of the bottle. Shall have sufficient holes to permit passage of functional parts through the cap.

3.2.3 Vent. Vent shall provide for passage of air from the atmosphere to the interior of the bottle.

* 3.2.4 Measuring and dispensing assembly. Measuring and dispensing assembly shall consist of a plunger and barrel assembly, a ball valve assembly, a delivery tube, and a scale with an adjustable indicator.

* 3.2.4.1 Plunger and barrel assembly. Plunger and barrel assembly shall be fabricated of glass. Top of plunger shall have a plastic or aluminum operating handle. When furnished, metering rod (Figure 2) shall be corrosion-resisting steel or glass-filled plastic.

3.2.4.2 Ball valve assembly. Delivery shall be accomplished by means of the action of two ball valves.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:56Z
Check the source to verify that this is the current version before use.

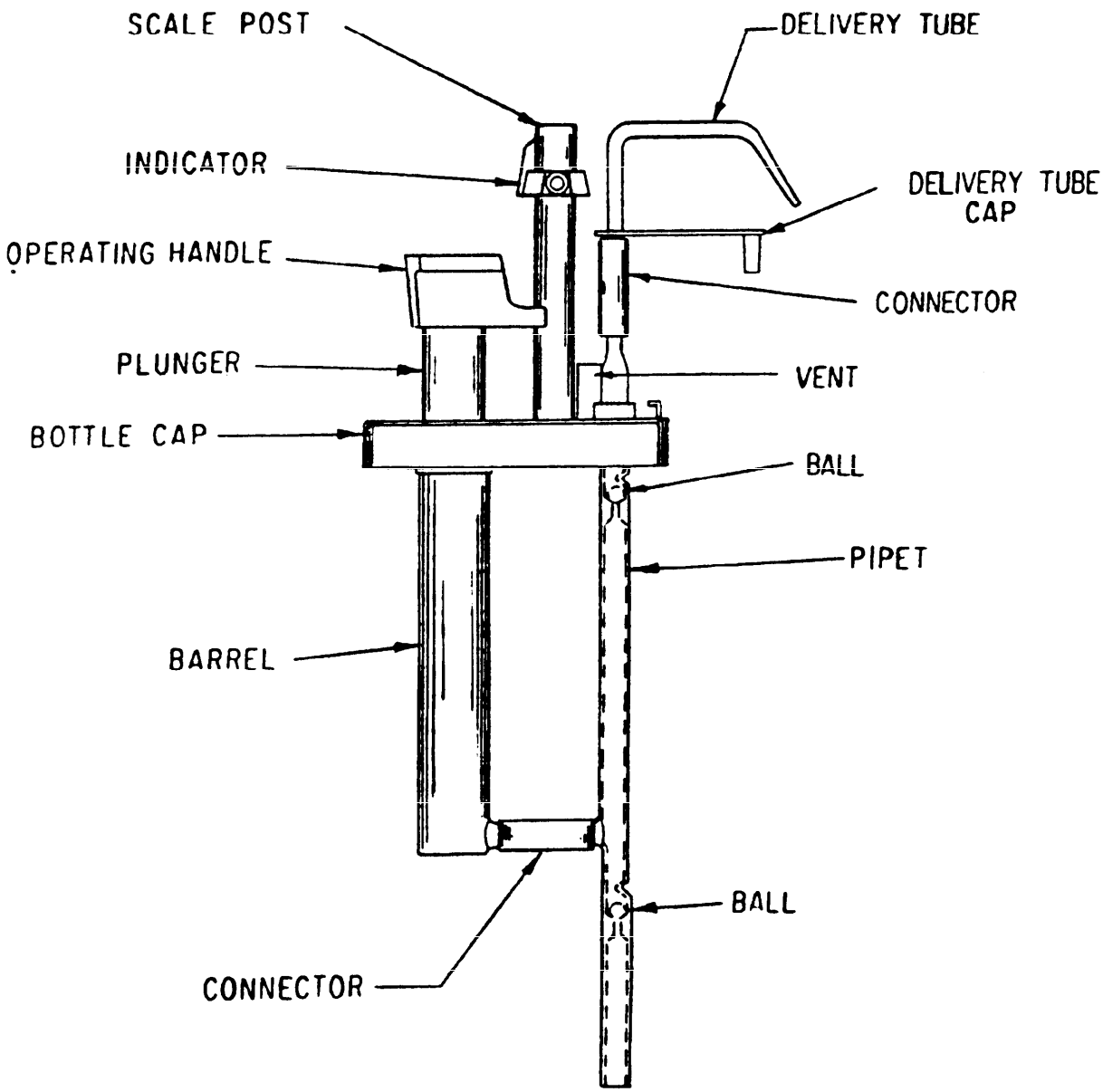


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

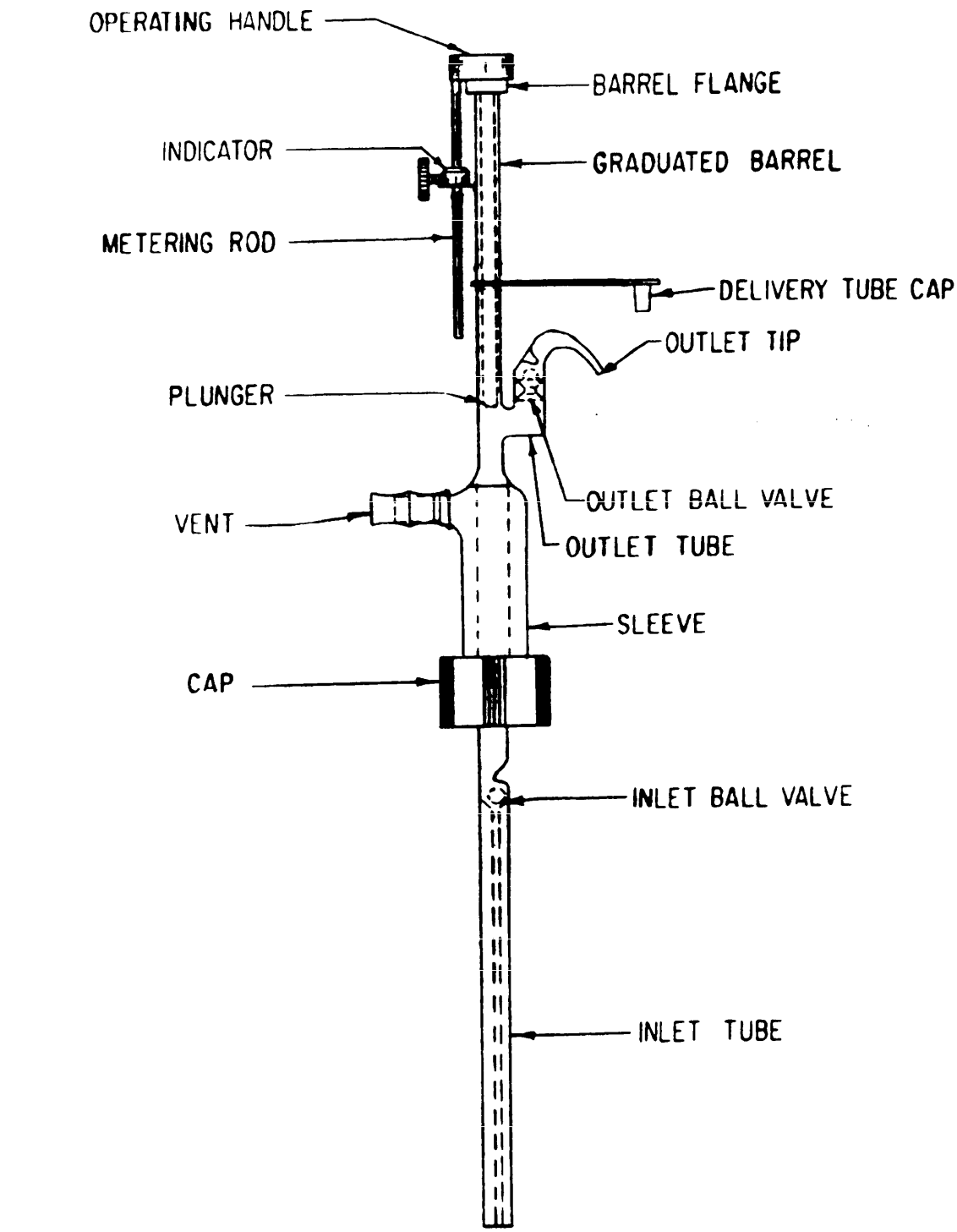


FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

3.2.4.3 Delivery tube. Delivery tube shall be fabricated of glass. A plastic cap which fits over the tip of the delivery tube shall be furnished. Cap shall be attached to the delivery tube by means of a flexible plastic ring and connector.

3.2.4.4 Indicator. Indicator shall be fabricated of plastic or aluminum. Means shall be provided for placing and securing the indicator at all applicable scale graduations specified in Table I.

* 3.2.4.5 Scale. Scale shall be marked on a separate plastic scale post (Figure 1) or on the exterior of the barrel or plunger (Figure 2).

3.3 Capacity and graduations. Bottle capacities shall be as specified in Table I. Graduation markings shall be as specified in Table I. Smallest graduation shall not exceed that shown in Table I.

TABLE I. Capacities and graduations

	Size 1	Size 2
Bottle capacity in ounces minimum	30	16
Graduated range in milliliters	0-10	0-1
Smallest graduation interval in milliliters	0.2	0.02
Numbered graduations in milliliters	0,1,2,3,4,5, 6,7,8,9,10.	0,.1,.2,.3,.4,.5, .6,.7,.8,.9,1.0

3.4 Performance.

3.4.1 Accuracy. Pipetting machine shall deliver the volume specified $\pm$ 2 percent when tested in accordance with 4.4.1.

* 3.4.2 Reproducibility. No delivery shall vary by more than 0.5 percent from the average delivery when calculated as specified in 4.4.2.

3.5 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review), and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.6 Marking.

3.6.1 Graduation markings. Graduation markings shall be permanent and legible.

3.6.2 Identification marking. Each pipetting machine shall be legibly marked with the name or registered trademark of the manufacturer.

3.7 Workmanship. Workmanship shall be first class throughout. Pipetting machines shall be free of defects listed in Table IV and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection. Inspection, as used in this specification is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105. As indicated in Table II, unit of product shall be one pipetting machine.

* TABLE II. Sampling for examination

For visual examination	Inspection level	AQL (percent defective)
Major defects	II	1.0
Minor defects	II	2.5

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and Table III. Unit of product shall be one pipetting machine, and acceptance number shall be zero.

TABLE III. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Accuracy	3.4.1	4.4.1	S-1
Reproducibility	3.4.2	4.4.2	S-2

4.3 Examination. The pipetting machine shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but shall not be limited to, the defects indicated in Table IV.

TABLE IV. Classification of defects

Category	Defect
Major	
101	Component missing.
102	Service data missing.
103	Component not free of breaks.
104	Component not free of cracks or chips.
105	Component not free of deformation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects (continued)

Category	Defect
Major	
106	Plunger does not move easily.
107	Indicator does not move easily. Indicator removes scale markings while being operated in accordance with instructions in service data.
108	Indicator cannot be secured.
109	Graduation marking not complete, correct, legible or permanent. Permanence shall be checked by rubbing scale markings vigorously with absorbent cotton which has been moistened with water. Entire range of scale shall be rubbed at least ten times.
Minor	
201	Pipetting machine not free of scratches.
202	Manufacturer's name or trademarks not complete, correct or legible.

* 4.3.2 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Accuracy. Reservoir shall be filled with distilled water at room temperature (22 $\pm$ 2°C.), and indicator shall be set at maximum scale graduation. Pipetting machine shall be operated in accordance with procedures specified in service data to deliver five samples. Volumes delivered shall be determined by gravimetric analysis. Accuracy shall be also determined at 3 ml. and 6 ml. volumes for Size 1 and .3 ml. and .6 ml. volumes for Size 2 pipetting machines.

* 4.4.2 Reproducibility. Reproducibility shall be determined from the five volumes determined in 4.4.1. Results shall be calculated as follows: Average delivery shall be determined. Variation from the average shall be determined for each delivery and shall be reported in percentage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

Example:

	ml. delivered	Variation from average in ml.	Percentage Variation
1.	10.01	.01	0.1
2.	10.05	.05	0.5
3.	9.95	.05	0.5
4.	10.01	.01	0.1
5.	9.98	.02	0.2

Average 10.00

* 5. **PACKAGING.**

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each pipetting machine , of the size specified, shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure of box shall be as specified in the appendix of PPP-B-636.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Nine pipetting machines, of the size specified, shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproof in accordance with PPP-B-636.

5.2.2 Level B. Nine pipetting machines, of the size specified, shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:56Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Dispensing of repetitive amounts of reagent solutions in laboratory procedures.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National stock number (NSN).
c. Size desired (see 1.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet military requirements.

6.4 This specification is applicable to the following items as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-176-5244	PIPETTING MACHINE, SEMI-AUTOMATIC, Manually Operated, 0-10 ml. Reagent Range.
6640-00-176-5245	PIPETTING MACHINE, SEMI-AUTOMATIC, Manually Operated, 0-1 ml. Reagent Range.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:56Z
Check the source to verify that this is the currenversion before use.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0947

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-P-36637B - PIPETTING MACHINE, SEMI-AUTOMATIC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:56Z
Check the source to verify that this is the current version before use.